

Calc is a new way of thinking
Calc2 builds on this ↑

Ø8.16.
2025

So what's Calculus?

Calculus is a mindset — a dynamics mindset. Contentwise, calculus is the mathematics of infinitesimal change!

Calculus as a way of thinking

Algebra + Geometry → largely focuses on static problems → Problems that lack change.

Where in Calc → change is central! about dynamics!

e.g. → Find the pattern in the seq
of numbers 2, 1, 1, 2, 3, 5, 8, 13
21 ... etc

Ø8.16.25

• While a calc question $\rightarrow$

Does the ratio of consecutive numbers
in the sequence above approach a specific number,
Fibonacci sequence $\rightarrow$ the n^{th} number
(denote that F_n) is the sum of the two
preceding numbers ($F_{n-1} \text{ \& } F_{n-2}$), starting
with $F_1 = 1$, $F_2 = 1$

A precalc (a static formula)

*** Calculus answer to the calculus question
Reveals something magical & enlightening

$$\lim_{n \rightarrow \infty} \frac{F_{n+1}}{F_n} = \varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618...$$

Golden Ratio

Golden Ratio

Ø8.16.25

↳ The ratio of consecutive Fibonacci numbers tends to φ (phi)

Furthermore $\rightarrow$ the sequence "Tends" $\rightarrow$
Further $\rightarrow$ into $\{ \text{dynamics} \} \rightarrow$ calc answers this!

Black Hole in the center of the Messier 87
 $\rightarrow$ constellation of Virgo

Golden Ratio $\rightarrow$ encoded in the spiral arrangements
of leaves on plant stems $\rightarrow$ proportions of components
in human hearts & Brains (what about Fractals?)
& theoretical models of Black Holes
' "Calc is hidden in you."

"Change is the mindset of Calculus"

Role of Infinitesimals in Calc $\phi 8.16.2\phi$
25

2, Zeno of Elea (ca. 490-430 BCE)
motion was not possible. $\rightarrow$ The Dichotomy Paradox

To travel a certain distance you must
first traverse half of it $\rightarrow$

$$d_n = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{2^{n-1}}$$

as Zeno continues his walk $\rightarrow$ The total distance
walked d_n always gets closer to 2 $\rightarrow$ but never
reaches 2 $\rightarrow$ Zeno's steps always traverses
half the remaining distance $\rightarrow$ Achilles & the tortoise

covers Infinite sum of the distances Zeno
with each step should equal 2

One of the pillars of Calc (1) provides
a stable foundation, on which to build
new concepts & methods $\rightarrow$

Limits!

8.16
2.5

Limits: Still $\rightarrow$ The foundation of Calc

the sum can be expressed as $\rightarrow$

$$d_n = 2 - \frac{1}{2^{n-1}} = 2 \left(1 - \frac{1}{2^n} \right)$$

$$\left[\lim_{n \rightarrow \infty} d_n = \lim_{n \rightarrow \infty} \left(2 - \frac{2}{2^n} \right) = 2 - \lim_{n \rightarrow \infty} \frac{2}{2^n} = 2 \right]$$

So, the two-foot mark is the limiting value of the total distance Zeno's traversing.

$\rightarrow$ is a statement about the dynamics of Zeno's walk

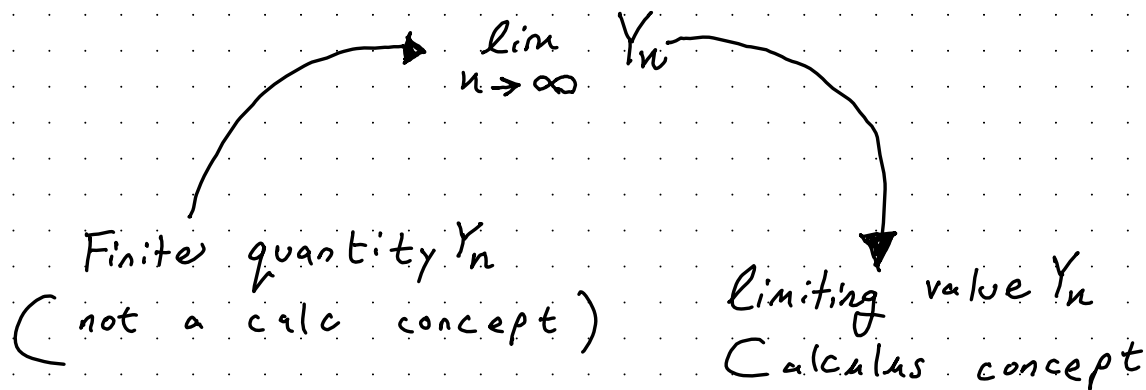
while the previous eq $\rightarrow$ is a static snapshots of How far Zeno has traveled!

ø8.16, 2ø25

after n steps

ϵ the 2nd eq reminds you that d_n is always
'approaching' 2 yet never arrives @ 2
from Calc 2 $\rightarrow$

Limits approach indefinitely
(ϵ thus never arrive!)



Limits are also the foundation of
Calc 2, Calc 1 is built on limits
 ϵ so will calc 2

Ø8.16.2025

: Start with a finite quantity

Y_n that depends on an integer n , & let n tend to infinity (i.e. $\rightarrow$ take the 'limit as $n \rightarrow \infty$ to arrive at a calculus results

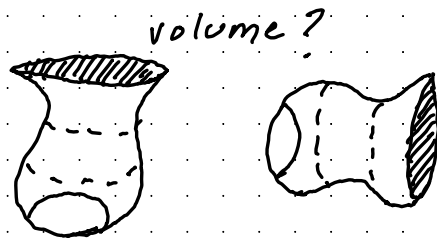
The 3 difficult Questions that drove the dev of Calc 2

Calc 2 developed out of the need to answer the following 3 Big Questions

1. Geometry Questions $\rightarrow$ Can you calc the length of any curve, area of any surface & volume of any solid? The anc Greeks + other civs could calc lengths, areas & volumes for Polyhedra & some curved shapes

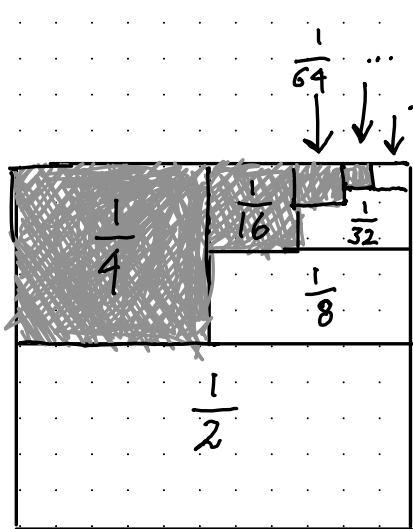
ø8.16.
2ø25

Spheres $\Rightarrow$ thousands of years
math bros struggled to calc the surface
areas & vols of objects as simple as a
flower vase 2,



volume?

surface area?



Length?

They could paint them
& fill them with the
water but not know
beforehand how much
paint or water they
need

The Infinite Sum Question

§8.16. 2025

Does an infinite sum have a sum, Σ if so, what's the sum?

For Centuries mathematicians have used Geometric args to tackle infinite sums

e.g. $\rightarrow$ adding all the areas inside the square

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots = 1$$

you multiply this equation by 2 which yields a 2nd (verification of that sum)

But does a general infinite sum $\rightarrow$ have a sum? * how can you calc that sum $\rightarrow$

3) The approx Question

Ø8.16.2Ø
25

Without knowing the exact value of a function
'2> you can accurately approx it?

Computers make it easy $\rightarrow$ like how to
approx quantities like $\sqrt{1.05}$ or other equations
like curves ... vases etc?

It took thousands of years

In the dynamic world $\rightarrow$ you fill beakers
with water '2> to measure volume $\hat{=}$ add up
numbers to obtain a sum!

look for the (Y_n)

ions use calc₂ to solve the proposed quest-

1 Think of Calc as a (verb) $\rightarrow$ action!
 $\hat{=}$ search for the finite quantities

that γ_n

ø6.18.
2ø25